BU MET CS544 A2 S2024 Exam Memorization Sheet **ANSWER KEY**

The topics to the left are the content/questions you should be able to describe/answer from memory (except for the formulas on the formula sheet). Complete the table by filling in the information, using your notes when you have to.

What are the three measures of center?	Mean, Median, Mode
What are the four measures of spread?	Standard Deviation, Variance, IQR, Range
When will the standard deviation be zero?	When all the values are the same
Interpret standard deviation	The typical distance from the data to the mean
Right Skewed vs Left Skewed	Right skewed has a tail on the right, with the mean larger than the median Left skewed has a tail on the left, with the median larger than the mean
Outlier Test	LB: Q1 – 1.5IQR UB: Q3 + 1.5IQR
Describe a Distribution	Shape, Outliers, Center, Spread
What are the summary statistics that are resistant to outliers?	Median, IQR
What are the summary statistics that are not resistant to outliers?	Mean, Standard Deviation, Range
percentiles	Percent below a certain point
z-score	$z = \frac{x-\mu}{\sigma}$

Standard Normal Distribution	N(0, 1)
What do the numbers 68-95-99.7 mean?	Empirical Rule; In a normal distribution, 68% of all data is within 1 standard deviation of the mean, 95% is within 2 standard deviations, and 99.7% is within 3 standard deviations.
What is the main difference between a segmented bar graph and a mosaic plot?	A mosaic plot takes the sample sizes of the explanatory variable into account by the distribution on the x-axis
Correlation Coefficient	r □ determines the strength of a linear relationship between x and y
What three things does the value of r communicate?	Linear form, direction, and strength
Coefficient of Determination	r^2 □ the percent of the variability in y that can be explained by the LSRL of y on x
Describe a Scatterplot	Form (linear, exponential), Association (pos/neg), Strength (strong/moderate/weak), and any outliers
Residual	Observed y – Predicted y
What does the LSRL minimize?	The square of the residuals – the vertical distances from the line to the points
Interpret the y-intercept	The predicted value of y when x is zero
Interpret the slope	For every 1 increase in x, we predict a [slope] increase in y.
What are the 4 sources of bias in a study?	Undercoverage, non-response, response bias, hidden bias

What are 2 biased sampling methods?	Voluntary response, convenience sample
What is the difference between a cluster sample and a stratified sample?	A cluster sample splits the population into heterogeneous groups and then an SRS is used to select clusters. A stratified sample splits the population into homogeneous groups and then an SRS is taken from each strata.
Why do we block in experiments?	It reduces variation in the treatment groups
Matched Pairs Experiment	All subjects receive both treatments
Why do we randomize in experiments?	It reduces the effect of variables that we may not know about
What is the main difference between an experiment and an observational study?	A treatment is applied in an experiment
What are the four principles of experimental design?	Randomization, Replication, Control, Comparison
What is the scope of inference for experiments and studies?	If the data was collected using random selection, you can make inferences about the population; if the data was analyzed using random assignment, you can make inferences for cause and effect.
What does it mean when two events are mutually exclusive?	They have no outcomes in common
General Addition Rule	P(A or B) = P(A) + P(B) – P(A and B)
General Multiplication Rule	Independent: P(A and B) = P(A)*P(B) Dependent: P(A and B) = P(A) * P(B|A)
What does it mean when two events are independent?	One event does not affect the probability of the other event

How can you test if two events are independent? (two ways)	P(A and B) = P(A)*P(B) P(A|B) = P(A)
Expected value of a discrete random variable	$\mu_X = E(X) = x_1 \bullet p_1 + x_2 \bullet p_2 + \ldots + x_n \bullet p_n$
Standard deviation of a discrete random variable	$\sigma_x = \sqrt{\Sigma\left(x_i - \mu_x\right)^2 \bullet p_i}$
Adding and subtracting random variables	The means get added or subtracted You only add the variances and take the square root to get the standard deviation
Linear transformation of a random variable	$E(a + bX) = a + bE(X)$ $Var(bX) = b^2 Var(x)$ $\sigma_{bX} = \|b\|\sigma_X$
What are the four conditions for a binomial setting?	Two possible outcomes Fixed probability Independent trials Fixed number of trials
Binomial Probabilities	binompdf(n, p, x) P(X = x) = $(n\ x)p^x(1 - p)^{n-x}$ binomcdf(n, p, x) P(X ≤ x) = P(X = 0) + P(X = 1) + …+ P(X = x)
Mean and standard deviation of a binomial random variable	E(X) = np $\sigma_X = \sqrt{np(1 - p)}$
What are the four conditions for a geometric setting?	Two possible outcomes Fixed probability Independent trials Trials until 1st success
Geometric Probabilities	geometpdf(p, x) P(X = x) = $(1 - p)^{x-1} \cdot p$ geometcdf(p, x) P(X ≤ x) = P(X = 1) + …+ P(X = x)
Mean and standard deviation of a geometric random variable	E(X) = $\frac{1}{p}$ $\sigma_X = \frac{\sqrt{1-p}}{p}$

Sampling distribution of a sample proportion	$\hat{p} \sim N\left(\mu_{\hat{p}}, \sigma_{\hat{p}}\right)$ $\mu_{\hat{p}} = p$ $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$
Sampling distribution of a sample mean	$\bar{x} \sim N\left(\mu_{\bar{x}}, \sigma_{\bar{x}}\right)$ $\mu_{\bar{x}} = \mu$ $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$
1-proportion z interval	$\hat{p} \pm z^{*}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
1 sample t-interval	$\bar{x} \pm t^{*}\frac{s}{\sqrt{n}}$
1-proportion z test	$z = \frac{\hat{p}-p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$
1 sample t test	$t = \frac{\bar{x}-\mu_0}{\frac{s}{\sqrt{n}}}$
When do we use t instead of z?	When conducting inference for means when we don't know σ and have to use S_x
Interpret p-value	The probability of getting the observed statistic, or more extreme, if H_0 is true.
What is a Type I Error?	Rejecting H_0 when you shouldn't
What is a Type II Error?	Not Rejecting H_0 when it is false
What is power?	Correctly rejecting H_0
P(Type I Error)	α

What happens to your margin of error as your confidence increases?	The ME gets larger; wider; bigger
What happens to your margin of error as your sample size increases?	The ME gets smaller; narrower
What happens to Type I error, Type II error, and power as α decreases?	Type I goes down Type II goes up Power goes down
How do you find the expected value on a two-way table for a chi-square test?	(row total)*(column total)/(table total)
How do you find the degrees of freedom in a two-way table?	df = (num of rows – 1)(num of columns – 1)
What is the main difference between a chi-square test for homogeneity and a chi-square test for association/independence?	A homogeneity test uses two separate samples from two unique populations while the association test uses one sample from a single population
What are the conditions needed for inference about a slope?	**L: Linear** check scatterplot for a linear relationship between x & y **I: Independent** if sampling w/o replacement, check n ≤ .1N **N: Normal** Histogram of residuals should be approx. Normal **E: Equal SD** Residual plot should have random scatter **R: Random** SRS or random assignment
t-interval for β	Used to capture β, the true population slope t* = invT(tail prob, df) df = n – 2 $b \pm t^{*} SE_{b}$ $SE_{b} = \frac{s}{s_{x}\sqrt{n-1}}$
t-test for β	$t = \frac{b}{SE_{b}} =$ ______ p-value = tcdf(L, U, df) df = n – 2